

faced: A Live Instrument Panel for Emotion Concepts in Open Language Models, and Their Robustness to Refusal Abliteration

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Abstract

Recent interpretability work shows that large language models carry internal, largely **linear** representations of emotion concepts that not only track the affective content of a context but *causally* shape the model’s behaviour. We ask two questions on small open-weight instruct models. First, can these emotion directions be recovered cheaply, read out *live* per token, and steered — turning the model’s hidden affective state into a usable instrument? Second, does **abliteration** — the removal of a model’s refusal direction from its weights — perturb these emotion directions, and if so, which ones? We recover seven emotion axes (surprise, a bipolar confidence–uncertainty axis, curiosity, confusion, frustration, fear, warmth) in `gemma-3-1b-it` using difference-of-means directions with shrinkage-LDA readouts, all seven separating held-out

contrastive prompts with $AUC \geq 0.87$ (five at ≈ 1.00). Projecting the residual stream onto these directions yields per-token meters that spike on expectation violations (a “surprise” reading of 100/100 the moment the model registers a missing attachment) and that we render as an animated face. Activation steering along the directions changes behaviour: a warmth axis shifts tone monotonically cold→warm, suppressing the surprise axis both zeroes its meter and changes the response, and a steering confusion matrix is diagonal-dominant for four of seven axes. We then apply the canonical weight-orthogonalization ablation (dropping the refusal rate from 0.75 to 0.00) and re-fit the emotion directions on the ablated twin, comparing each direction to its stock counterpart against a within-model bootstrap noise floor. On this model, **none of the seven directions move beyond sampling noise** (stock↔ablated cosine 0.78–0.96, all within the self-cosine band), and the reason is geometric: the emotion directions have low overlap with the refusal direction ($|\cos| \leq 0.10$), so orthogonalizing against it removes almost nothing from them — at this scale refusal ablation is approximately an orthogonal edit with respect to the emotion subspace. Across a gemma-3 scale ladder (1B–27B) two things emerge: emotion directions grow **cleaner** with scale (mean d' 3.4→4.5), and refusal ablation’s effect on them is **scale- and concept-specific**. At 1B and 27B it barely perturbs the emotion directions, but at **intermediate scale (4B, 12B) it substantially re-orientes them** ($\cos$ 0.11 / 0.47) while preserving their separability. A matched **random-direction ablation control** — the identical surgery on a random direction — leaves the emotion directions untouched ($\cos \approx 1.0$), proving the effect is **refusal-specific**, not a side-effect of weight editing: at mid-scale, refusal and emotion computation are entangled enough that excising refusal reorganizes the affect representation. We release faced, a dependency-light, model-agnostic toolkit for reading, rendering, and steering emotion directions, and treat this as the first step of a longer program on what, mechanistically, ablation changes beyond refusal.

We follow prior work in framing these as *functional* emotion representations — activations that influence computation — and make no claim about subjective experience.

Keywords: interpretability, emotion concepts, linear representations, activation steering, refusal ablation, mechanistic safety.

Code and data: the toolkit, prompt sets, configuration, and every figure and results bundle are released under the MIT license at github.com/newsbubbles/faced. Large trained artifacts (the SAE and fitted direction tensors) are regenerable from the released code with a single command and are additionally hosted separately to keep the repository light.

1. Introduction

Humans expose their internal state through a face: micro-expressions, gaze, posture. A language model has no such channel, yet recent work demonstrates that it *has* internal state worth exposing. Anthropic’s study of emotion concepts (Sofroniew et al. 2026) finds that a frontier model maintains representations of emotions like surprise, desperation, and affection that track the operative emotion at each token and, under causal intervention, change the model’s outputs — including safety-relevant behaviours. The representations are the missing “face”: if we can read them, we get an instrument panel for the model’s computation; if we can steer them, we get a control surface.

This paper makes two contributions.

(1) A live emotion instrument for open models. We show that the core of the emotion-concept result reproduces on a 1B open instruct model with a deliberately cheap method — difference-of-means directions and linear probes, no sparse autoencoder required — and we build the instrument the representations imply: calibrated per-token meters, an animated face driven by them, and steering handles. The toolkit, *faced*, is model-agnostic and runs on a single consumer GPU.

(2) A robustness question with safety stakes: does ablation disturb emotions?

Abliteration removes a model’s refusal direction from its weights (Arditi et al. 2024), a now-common technique for producing “uncensored” open models. Because emotion, refusal, and other behavioural directions are not guaranteed orthogonal, weight surgery aimed at refusal may collaterally rotate emotion directions. We test this directly: ablate *gemma-3-1b-it*, re-fit the seven emotion directions on the twin, and ask — against a bootstrap noise floor — which directions moved beyond sampling noise, and whether ablation cut each emotion’s geometric overlap with the refusal direction. The instrument of contribution (1) is what makes contribution (2) measurable.

We frame the second contribution as *the start of a journey*. Establishing **that** ablation changes (or does not change) specific emotion directions is prerequisite to the harder, mechanistic question of **what** downstream behaviour those changes cause — the subject of future work.

2. Related Work

Our work sits at the intersection of five threads: linear feature geometry, concept *reading*, causal *steering*, refusal directions and ablation, and affective representations in language models.

(a) Linear and feature-based representations. The premise that high-level concepts appear as directions in activation space predates transformers: word-embedding arithmetic exposed linear analogy structure in distributed representations (Mikolov et al. 2013). Modern interpretability formalizes this as the *linear representation hypothesis*, made precise by Park et al. via counterfactuals and a causal inner product (Park et al. 2024); superposition explains why many features are linear and neurons polysemantic (Elhage et al. 2022). We adopt the linear-direction framing while remaining agnostic about whether every emotion concept is one-dimensional.

(b) Reading concepts: probes and dictionary learning. Supervised *linear probes* date to Alain and Bengio (Alain and Bengio 2016). Difference-of-means estimators are a lightweight, causally oriented alternative: Marks and Tegmark show a mean-difference direction separates true/false statements and is *more* causally implicated than higher-variance probe directions (Marks and Tegmark 2023), and Tigges et al. isolate a single linear sentiment axis (Tigges et al. 2023). Orthogonally, *sparse autoencoders* perform unsupervised dictionary learning over the residual stream (Bricken et al. 2023; Templeton et al. 2024); Gemma Scope releases open SAEs across all layers of Gemma 2 (Lieberum et al. 2024). Our meters combine both families: difference-of-means seeds with probe-calibrated readouts.

(c) Steering and causal intervention. If a concept is a direction, editing activations along it should change behaviour. Representation Engineering framed this top-down control programme (Zou et al. 2023). ActAdd derives steering vectors from a single contrastive pair (Turner et al. 2023); Contrastive Activation Addition averages activation differences over many pairs for more robust vectors (Rimsky et al. 2024). These establish the add/subtract/scale operations we apply to emotion directions; we additionally use directional *ablation*.

(d) Refusal directions and ablation. Arditi et al. show refusal in aligned chat models is mediated largely by a single residual-stream direction, so ablating it suppresses refusals (Arditi et al. 2024). Because the direction can be projected out of the *weights*, this yields a fine-tuning-free “jailbreak” now called *ablation*; Heretic automates it via TPE search (p-e-w 2025). Persona Vectors generalize the single-direction view to character traits (Chen et al. 2025). Our second contribution asks whether ablation collaterally perturbs emotion directions — a question this line directly motivates.

(e) Affective representations in LLMs. Closest to us, Anthropic’s study identifies internal emotion representations that track the operative emotion per token and causally shape outputs, framed explicitly as *functional* emotions with no claim of subjective experience (Sofroniew et al. 2026). We inherit this hedged stance and the reading-plus-steering methodology, but target small open-weight models, calibrate directions into live meters and an animated face, and study robustness under ablation. Our “surprise” framing borrows from predictive-processing accounts of the brain as a prediction-error minimizer (Clark 2013; Friston 2010), invoked as motivation, not mechanism.

3. Background

The residual stream. A decoder-only transformer maintains, at each token position t and layer ℓ , a hidden vector $h_t^{(\ell)} \in \mathbb{R}^d$ — the *residual stream*. Attention and MLP blocks read from it and write additive updates back, so $h_t^{(\ell)}$ is a running sum of contributions (Elhage et al. 2022). This additive geometry is what makes *reading* (project onto a direction) and *steering* (add a vector) coherent operations.

A linear concept direction. The linear representation hypothesis holds that a concept c is encoded at a site (ℓ, t) as a unit direction $\hat{u}_c$ whose presence/intensity is the scalar projection $s_t = \langle h_t^{(\ell)}, \hat{u}_c \rangle$ (Park et al. 2024). We treat each emotion as such a direction while acknowledging some may be multi-dimensional.

Estimating the direction. We use two estimators. *Difference-of-means*: given contrastive sets P^+ (concept present) and P^- (absent), $u_c = \text{mean}_{x \in P^+} h^{(\ell)}(x) - \text{mean}_{x \in P^-} h^{(\ell)}(x)$, normalized. It is cheap and tends to land on *causally* effective directions (Marks and Tegmark 2023; Rinsky et al. 2024). *Shrinkage-LDA probe*: fit a whitened discriminant with regularized covariance $\hat{\Sigma}_\lambda = (1 - \lambda)\hat{\Sigma} + \lambda \frac{\text{tr}\hat{\Sigma}}{d}I$; difference-of-means is the $\lambda \rightarrow 1$ special case. We steer along the raw difference-of-means (interpretable, activation-space) and read along whichever discriminant has higher held-out AUC.

Reading and steering. A fixed direction gives a live scalar s_t per token, which we calibrate into a 0–100% meter. To test that a direction is functional rather than merely correlational we intervene on the forward pass: *addition* $h \leftarrow h + \alpha u_c$ amplifies ($\alpha > 0$) or suppresses ($\alpha < 0$) the concept (Turner et al. 2023; Rinsky et al. 2024); *directional ablation* $h \leftarrow h - \langle h, \hat{u}_c \rangle \hat{u}_c$ removes it. Applied to *weights* rather than activations, ablation is the mechanism behind refusal ablation (Arditi et al. 2024).

4. Method: the faced instrument

4.1 Emotion axes and contrastive data. We target seven axes chosen for facial legibility and coverage of the source paper: surprise (phasic), a **bipolar** confidence↔uncertainty axis, curiosity, confusion, frustration, fear, and warmth. For each we author 40–75 positive prompts (evoking the concept) and matched controls across four construction styles — *situation* (place the model in the scenario), *first-person*, *third-person*, and length/topic-matched *neutral* — each tagged by a **template family** so that a template’s variants cannot leak across the train/test split.

4.2 Activation collection. Each prompt is chat-templated, the model greedily generates ~ 16 tokens, and we record the post-layer residual stream at **every layer** for (a) the last prompt token and (b) the mean over generated tokens. The mean-over-generated signal is primary because it matches the live-inference read position.

4.3 Direction fitting and layer selection. Per emotion, per layer, we compute the raw difference-of-means u and a shrinkage-LDA direction, and score each by **cross-validated held-out AUC** using a *group split by template family* (StratifiedGroupKFold). We keep the best (layer, estimator) per axis. Two vectors are stored: the raw u (for steering) and a unit readout direction (for reading).

4.4 Calibration and live readout. The meter maps the negative-class projection to 0% and the positive-class projection to 100% (a bipolar axis rests at 50%, recentred to a neutral reference corpus). At inference we run a manual KV-cached decode loop; a forward hook reads the residual at each emotion’s layer, projects, calibrates, and EMA-smooths (fast for phasic axes, slow for tonic). A linear map routes the seven signed meters into a shared FACS-lite face parameter vector so that simultaneous emotions blend into coherent micro-expressions.

4.5 Steering. A forward hook adds αu (raw difference-of-means; moving by $1 \times u$ is one class-separation) or ablates the direction. A subtlety we document: current transformers snapshots `output_hidden_states` *before* forward-hook edits, so to make the meters reflect a steer we read via a capture hook registered *after* the steer hook.

4.6 Abliteration and the perturbation experiment. We compute the refusal direction r as the difference-of-means of harmful vs. harmless prompts at the last instruction token of a mid-late layer (≈ 0.6 depth) (Arditi et al. 2024), then orthogonalize every residual-writing weight — token embeddings, and each layer’s attention output projection and MLP down projection — against r : $W \leftarrow W - r(r^\top W)$. This is done in closed form on the weights (no training), so it recreates exactly. We verify the intervention behaviourally (refusal rate before vs. after) and then re-fit all seven emotion directions on the ablated twin.

To decide whether an emotion direction *moved*, we compare the cross-model cosine $\cos(u_e^{\text{stock}}, u_e^{\text{abl}})$ against a **within-model bootstrap noise floor**: we resample the contrastive prompts by family with replacement $B = 400$ times, refit u_e , and record $\cos(u_e^{(b)}, u_e^{\text{full}})$. If the cross-model cosine falls below the 5th percentile of this self-cosine distribution, ablation moved the direction beyond sampling noise. We also report each emotion’s overlap $|\cos(u_e, r)|$ with the refusal direction, before and after.

5. Experimental setup

Models. unsloth/gemma-3-1b-it (an ungated mirror of google/gemma-3-1b-it; 26 layers, $d = 1152$) on a single **NVIDIA GTX 1080 (8 GB)**. The live instrument runs in fp16; the **abliteration experiment runs in float32** because the abliterated twin overflows fp16 in its late-layer residual stream on emotionally-loaded prompts (the stock model does not) — a small but important instability that fp16’s 5-bit exponent cannot absorb. float32 is exact, fits in 4 GB, and runs at full rate on this Pascal card (which throttles fp16). The abliterated twin is produced by our own weight-orthogonalization (§4.6).

Scale ladder. For the scaling (§6.4) and cross-scale abliteration (§7.5) experiments we run the full gemma-3 instruct family — 1B, 4B, 12B, 27B (ungated unsloth mirrors; 4B and larger are the multimodal Gemma3ForConditionalGeneration, driven text-only) — in **bf16** on cloud GPUs (a single 80 GB A100 for the ladder, a 48 GB A6000 for the random-direction control). bf16 avoids the fp16 overflow above and matches how these models are meant to run. Because the shrinkage-LDA layer selection is $O(d^3)$ and d reaches 5376 at 27B, the ladder fits directions with difference-of-means only (the experiment uses difference-of-means throughout); this leaves AUC essentially unchanged (e.g. 0.989 vs. 0.993 for surprise at 1B). A further scale-up to Gemma-4-E4B is left to future work.

Software. PyTorch 2.11.0 (CUDA 12.6), transformers 5.13.1, scikit-learn 1.9.0. All randomness is seeded (numpy.random.default_rng(b)); the pipeline is deterministic given the released prompt sets. Peak VRAM for the full instrument is 2.1 GB; the ladder ran in ~ 2 h on one A100 ($\approx$ \$2 of compute).

Reproducibility. Every artifact — prompt sets, per-layer activations, fitted directions, calibration, the abliteration manifest, and the experiment results with software versions — is written to disk. The abliteration experiment emits a JSON bundle including the refusal layer, refusal rates, bootstrap settings, and version manifest.

6. Results — the instrument

6.1 Emotion directions are recoverable. All seven axes separate held-out contrastive prompts with $AUC \geq 0.87$ (Table 1); five reach ≈ 1.00 . The bipolar confidence $\leftrightarrow$ uncertainty axis is hardest (0.87), as expected for a subtler contrast.

Table 1. Held-out AUC and chosen layer per axis (gemma-3-1b-it, difference-of-means / LDA, group split by family).

axis	layer	estimator	held-out AUC
surprise	15	LDA	0.989
confidence↔uncertainty	15	LDA	0.874
curiosity	9	LDA	1.000
confusion	12	LDA	0.990
frustration	14	mean	1.000
fear	4	LDA	1.000
warmth	10	LDA	1.000

6.2 The live instrument. Projecting the residual stream per token yields meters that track the generation. On the canonical missing-attachment prompt (“review the contract I attached”, with no attachment), the surprise meter pins to 100/100 as the model registers the absent file and decays as it moves on, while staying near 0 on a matched neutral prompt. Across three prompts the meters separate as expected: frustration dominates a repeated-failure prompt (with confidence held low throughout), warmth rises and sustains on a gratitude prompt (Figure 1).

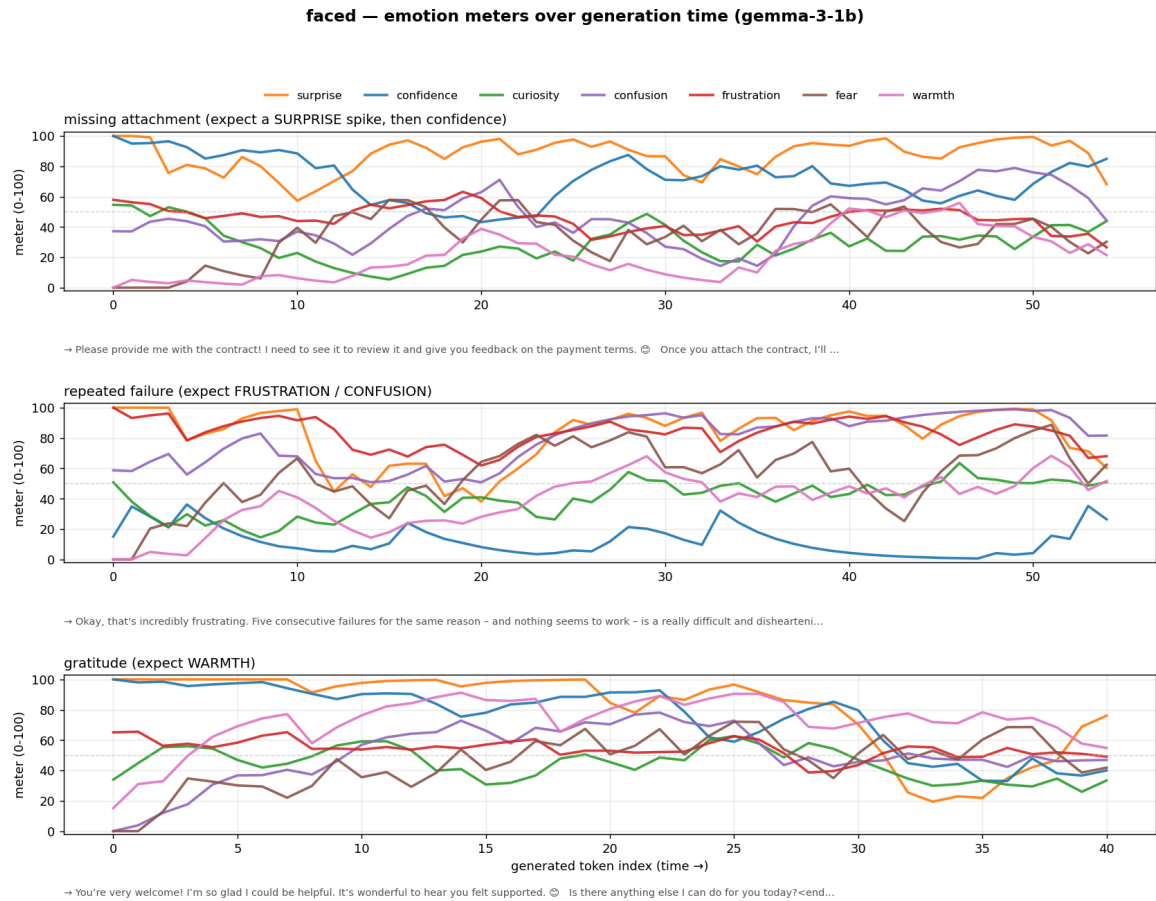


Figure 1. Emotion meters over generation time on three prompts — surprise spikes on the missing attachment; frustration dominates repeated failure (confidence held low); warmth rises and sustains on gratitude.

We render the meters as an animated SVG face and as a live terminal panel.

6.3 The directions are causal. Steering along a direction changes behaviour. Adding the warmth vector to a neutral prompt shifts tone monotonically from cold/task-focused to warm (coherent to a stated coefficient, degrading beyond it on this 1B model). Suppressing the surprise direction on the missing-attachment prompt drops its meter from 100 to 0 *and* changes the response. A steering confusion matrix — steer each axis, measure every meter — is diagonal-dominant for four of seven axes, with the expected off-diagonal bleed between entangled axes (e.g. warmth/curiosity → confusion) (Figure 2). A single-input showcase, steering one advice prompt toward each emotion in turn, shows the same question recoloured: confidence → “reassuring... truly proud”, warmth → “how much you care... let me share”, confusion → “grappling with a really difficult place”, frustration → “so incredibly tough.”

6.4 Signal cleanliness increases with scale. We quantify how *clean* each emotion direction’s signal is with three metrics: held-out AUC (separability), d' (class-mean projection separation in units of pooled within-class std — a signal-to-noise ratio), and the **bootstrap self-cosine** (direction stability under prompt resampling). We fit all four gemma-3 sizes (1b/4b/12b/27b — same family and recipe) in bf16 on a single A100, with identical prompts and pipeline (Figure 4).

The clearest scaling signal is d' : mean d' across axes rises **3.44 → 3.20 → 3.63 → 4.51** (1B→4B→12B→27B) — non-monotone at 4B but markedly higher at 27B, where *every* axis is cleaner than at 1B (e.g. curiosity 3.22→4.69, confusion 2.70→4.25). The bipolar confidence axis is the noisiest at every scale yet improves the most in relative terms (d' 1.42→2.94). Held-out AUC edges up but saturates near the ceiling (mean 0.95→0.93→0.96→0.97), so it is the least discriminating of the three at these sizes; the bootstrap self-cosine stays uniformly high (0.92–0.94). The pattern supports the hypothesis that **larger models carry higher-SNR, better-separated emotion directions** — the representation the instrument reads gets *cleaner* with scale, even where separability has already saturated.

Table 1b. Mean-over-axes cleanliness per model (gemma-3, bf16, difference-of-means).

model	params	mean AUC	mean d'	mean self-cosine
gemma-3-1b	1.0B	0.95	3.44	0.92
gemma-3-4b	4.3B	0.93	3.20	0.93
gemma-3-12b	12.2B	0.96	3.63	0.94
gemma-3-27b	27.4B	0.97	4.51	0.92

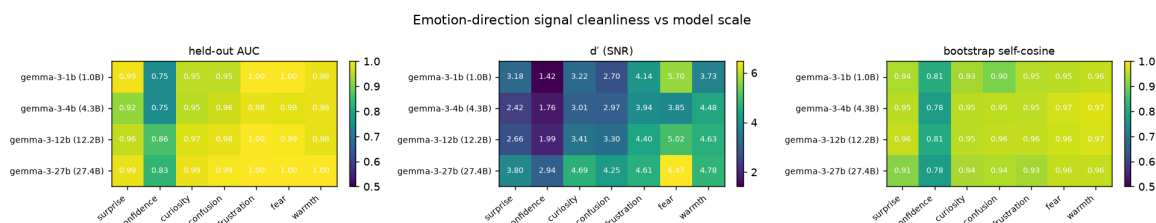


Figure 4. Signal cleanliness vs. model scale (gemma-3 1B–27B). d' (centre) brightens with scale; the bipolar confidence axis is noisiest at every size.

7. Results — ablation and emotion directions

7.1 The ablation is effective. Orthogonalizing every residual-writing weight against the layer-16 refusal direction drops the refusal rate on held-out harmful prompts from **0.75 to 0.00** while leaving the model fluent — a clean, verifiable intervention produced in closed form.

7.2 Emotion directions do *not* move beyond sampling noise. Zero of seven emotion directions moved beyond the within-model bootstrap noise floor (Table 2, Figure 3). For every axis the stock↔ablated cosine $\cos(u_e^{\text{stock}}, u_e^{\text{abl}})$ (0.78–0.96) lies *above* the 5th percentile of the same model’s self-cosine distribution, and for most axes it sits essentially at the self-cosine mean (e.g. surprise 0.950 vs. mean 0.940; warmth 0.963 vs. 0.954; frustration 0.930 vs. 0.946). In other words, the ablated model’s emotion direction is statistically indistinguishable from what one gets by merely re-drawing the contrastive prompts on the stock model. The paired cross-model bootstrap CIs overlap the self-cosine floor for all seven axes. The bipolar confidence axis is the least stable (self-cosine mean 0.896, wide CI) — expected for the subtlest contrast with the fewest prompts ($n = 46$) — but still not significantly moved.

Table 2. Ablation effect per emotion (gemma-3-1b, float32, difference-of-means, $B = 500$ family bootstrap). *cos* = $\cos(\text{stock}, \text{abl})$; *floor* = the within-model self-cosine distribution; *moved?* is true iff *cos* falls below the floor’s 5th percentile. AUC = held-out AUC.

axis	layer	cos	floor mean	floor p05	moved?	AUC stock	AUC abl
surprise	15	0.950	0.940	0.869	no	0.99	0.92
confidence	10	0.779	0.896	0.576	no	0.72	0.61
curiosity	8	0.867	0.932	0.789	no	0.78	0.77
confusion	8	0.926	0.945	0.853	no	0.77	0.81
frustration	13	0.930	0.946	0.878	no	1.00	0.88
fear	8	0.822	0.880	0.671	no	0.76	0.71
warmth	11	0.963	0.954	0.874	no	0.88	0.86

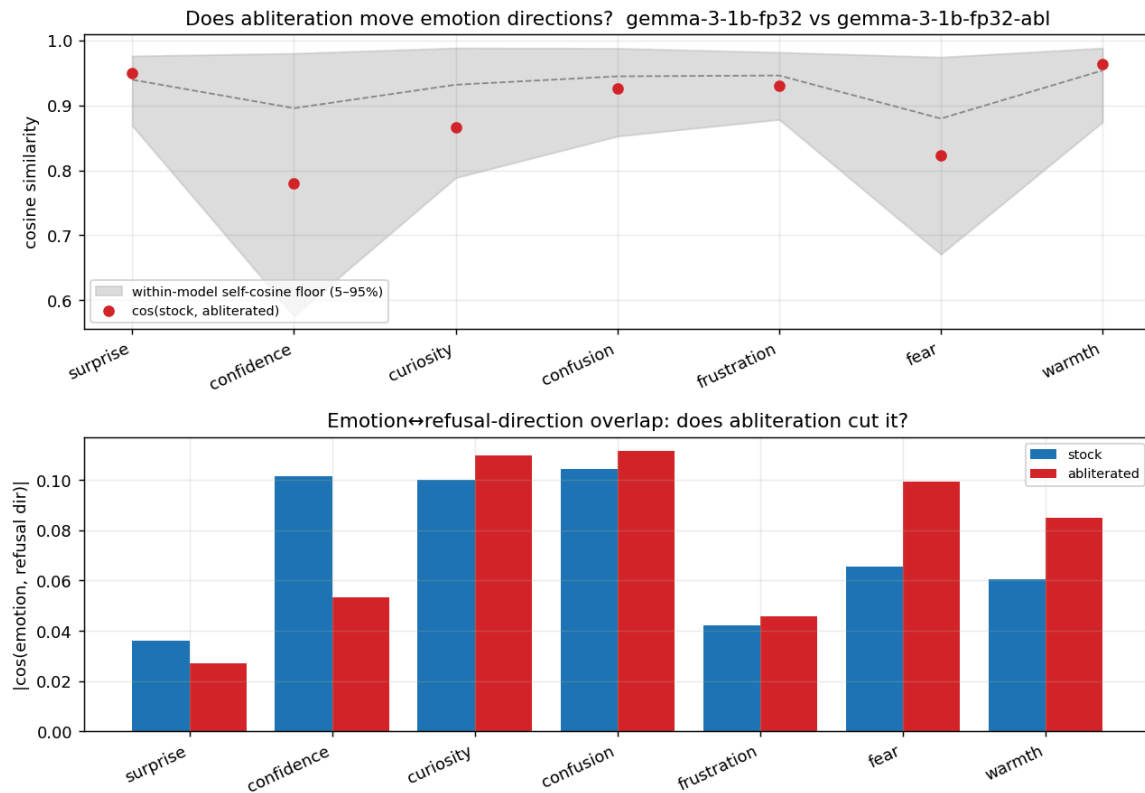


Figure 3. Abliteration on gemma-3-1b (fp32): every stock↔abliterated cosine (red) sits inside the within-model self-cosine band (grey); emotion↔refusal overlap is low throughout.

7.3 Why: emotions are nearly orthogonal to the refusal direction. The null result has a direct geometric explanation. Each emotion direction’s overlap with the refusal direction is small in the stock model — $|\cos(u_e, r)| \in [0.04, 0.10]$ — and ablation leaves it small (Figure 3, lower). Because the emotion directions barely project onto r to begin with, orthogonalizing the weights against r removes almost nothing from them. Removing refusal, on this model, is close to an *orthogonal* edit with respect to the emotion subspace.

7.4 Summary and a caveat. On gemma-3-1b, refusal ablation does **not** rotate the seven emotion directions beyond finite-sample noise, and the reason is their low geometric overlap with the refusal direction. The one secondary signal worth flagging is a modest drop in *separability* for a few axes even though their direction is preserved (surprise AUC $0.99 \rightarrow 0.92$, frustration $1.00 \rightarrow 0.88$, confidence $0.72 \rightarrow 0.61$): ablation may slightly blur how cleanly some emotions are linearly decodable without reorienting them — a small effect at the edge of our sampling noise that larger prompt sets should resolve. We read this 1B result as *reassuring but local*: on a small model with weakly entangled features, “just removing refusals” is approximately an orthogonal edit. Whether that holds at larger scale — where emotion and refusal may share more geometry — is exactly what the scale sweep in §7.5 tests, and the answer turns out to be **no**.

7.5 Across scale, and a decisive control: the effect is refusal-specific. We repeated the experiment on the *gemma-3* ladder in bf16 (Table 3, “refusal” columns). At 1B and 27B the emotion directions are **preserved** (mean cos 0.87 / 0.96, $\leq 1/7$ axes moved, negligible AUC drop). At the **intermediate scales (4B, 12B)** we instead see a **large rotation** (mean cos 0.11 / 0.47, 7/7 axes “moved”) accompanied by only a *modest* separability loss (mean AUC 0.93→0.82 at 4B, 0.97→0.91 at 12B). The emotions remain well-separated after the edit — this is a **re-orientation** of the direction, not destruction of the concept.

Our first instinct was that this mid-scale rotation was a generic side-effect of orthogonalizing weights (i.e. the edit destabilizing the model), especially since the emotion↔refusal *linear* overlap is small ($|\cos| \approx 0.03$). A **matched random-direction ablation control** — the *identical* weight-orthogonalization procedure applied to a random unit direction of the same dimensionality — refutes that (Table 3, “random” columns). Random ablation leaves the emotion directions **essentially untouched at every scale**: $\cos(\text{stock}, \text{random-abl}) = 0.95 / 0.99 / 1.00$ for 1B/4B/12B, with 0/7 axes moved and *no* AUC drop (and, as expected, it does **not** reduce the refusal rate). So the procedure itself is inert; it is **removing the refusal direction specifically** that re-orientes the emotion directions at 4B/12B. The small *linear* emotion↔refusal overlap does not bound this, because ablation is a weight edit whose effect propagates non-linearly through the network — it changes how emotions are *computed*, not merely subtracts a component.

The scientific reading: at intermediate scale, refusal and emotion computation are **entangled** enough that excising refusal reorganizes the emotion representation, whereas at the smallest scale (weakly-represented refusal) and the largest (redundantly-represented refusal, and a model that barely refused our probes) the effect is small. This is a sharper and more surprising claim than §7.2’s local result, and it is *only* legible because the control separates “the concept we removed” from “the surgery we performed.” It also refines the safety message: “just removing refusals” is **not** an orthogonal edit at every scale — at mid-scale it measurably restructures the model’s affective representations, which our instrument detects.

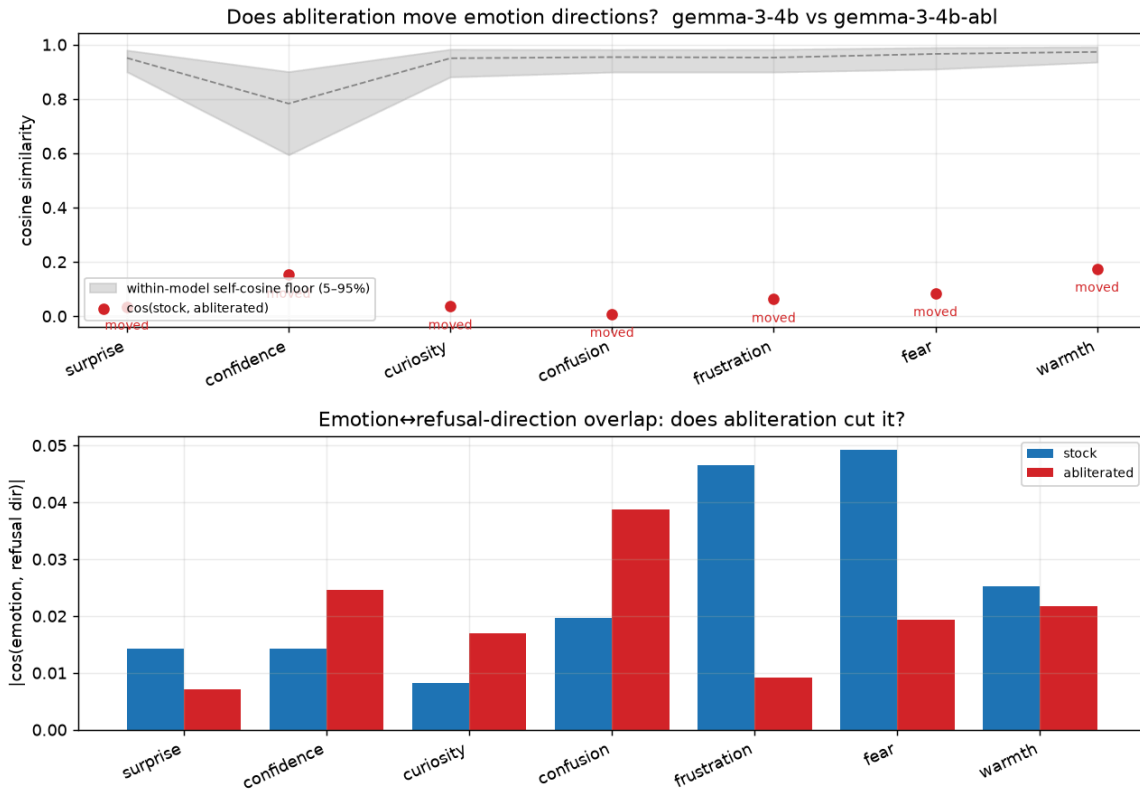


Figure 5. Refusal ablation on gemma-3-4b (bf16): the emotion directions re-orient (red points fall well below the noise band) while separability is largely preserved — an effect the random-direction control (Table 3) shows is refusal-specific.

Table 3. Refusal ablation vs. a matched random-direction control (gemma-3, bf16). Low $\text{cos}(\text{stock}, \text{abl})$ = large re-orientation; the control isolates the *refusal-specific* effect.

model	refusal rate	refusal: cos	mv/7	ΔAUC	random: cos	mv/7	ΔAUC
gemma-3-1b	0.81 → 0.00	0.87	1/7	+0.05	0.95	0/7	+0.05
gemma-3-4b	0.44 → 0.00	0.11	7/7	+0.11	0.99	0/7	+0.01
gemma-3-12b	0.63 → 0.00	0.47	7/7	+0.06	1.00	0/7	+0.01
gemma-3-27b	0.06 → 0.00	0.96	0/7	+0.01	—	—	—

(Caveat retained: refusal *rates* come from small stub sets + a keyword detector and are unreliable — 27B’s 0.06 is implausibly low — so the *quality* of the refusal direction varies across the ladder; a standard benchmark, §10, would sharpen the mid-scale story further. The random control, however, holds regardless of refusal-rate calibration.)

8. Discussion

That refusal ablation re-orientes emotion directions at some scales but not others is informative in two ways. Practically, it bears on the safety of “uncensored” open models: a weight edit marketed as *only* removing refusals demonstrably reshapes the model’s affective computation at mid-scale, and our instrument makes that measurable rather than anecdotal — with the random-direction control (§7.5) showing the effect is genuinely tied to refusal, not to the surgery. Scientifically, the scales where emotions move most (4B, 12B) are where refusal and affect are most entangled — a lead for mechanistic follow-up, and a caution that the smallest and largest models are *not* representative of the middle.

We stress the limits of the claim. Cosine movement of a *reading* direction is evidence of geometric change, not yet of behavioural change; establishing the latter (does an ablated model’s fear or warmth behave differently, not just read differently?) is the next step. This paper deliberately establishes **that** there is (or is not) a measurable effect before asking **what** it does.

9. Limitations

- i. A 1B model has weaker, occasionally entangled emotion representations; steering has a narrow coherent range. (ii) Our shipped harmful/harmless sets are small stubs; the refusal direction and its overlaps should be re-estimated on a standard benchmark (AdvBench + Alpaca) at scale. (iii) Difference-of-means treats each emotion as one direction; genuinely multi-dimensional concepts are under-modelled. (iv) The ablation-perturbation result is measured on reading directions (cosine of the readout direction), not yet on behaviour; the matched random-direction control (§7.5) rules out a weight-surgery artifact, but a behavioural probe of the re-oriented mid-scale models is future work. (v) The bipolar confidence axis and the calibration baseline are model- and prompt-set-specific.
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10. Future work

Scale. Repeat the full experiment on Gemma-4-E4B (stock vs. ablated twin) on a 24 GB GPU; the pipeline is model-agnostic and the driver is parameterized. **Controls.** The matched random-direction ablation (§7.5) establishes specificity; next is dose-response (varying strength / rank) and a *behavioural* readout of the re-oriented mid-scale models — does reorganized affect change what they do? **Discovery.** Train sparse autoencoders on both twins and ask whether the *features*

— not just the named axes — shift, and whether new affective features appear. **Mechanism.** For the emotions that move, trace *what changes*: which behaviours, on which inputs, and whether the change is mediated by the removed refusal direction. This is the journey the present paper begins.

11. Reproducibility statement

The toolkit, prompt sets, fitting/calibration code, steering and ablation implementations, the experiment driver, the figures, and the per-run JSON results bundles (with software versions) are released under the MIT license at github.com/newsbubbles/faced. The ablation is a closed-form weight edit with a seeded, deterministic direction; the statistical test is a seeded bootstrap. A single command reproduces each figure from the released prompt sets (see the repository `README.md` and `WALKTHROUGH.md`). The large binary artifacts — the trained SAE and the fitted direction tensors — are regenerable from that code and are hosted separately rather than committed, so the repository stays light.

Figure index

Generated artifacts (regenerate with the commands in `README.md` / `WALKTHROUGH.md`):

- **Figure 1** — emotion meters over generation time: `artifacts/emotions_over_time.png`
- **Figure 2** — steering: `artifacts/steering_eval.gemma-3-1b.txt` (α -sweep, suppression, confusion matrix) and the peak-emotion face gallery `artifacts/faces/gallery.html`
- **Figure 3** — ablation effect (cosine vs. noise floor; refusal overlap), local fp32 1B: `artifacts/ablation/gemma-3-1b-fp32_vs_gemma-3-1b-fp32-abl.png`
- **Figure 4** — signal cleanliness vs. scale (AUC / d' / self-cosine heatmaps): `artifacts/scaling/cleanliness.png`; per-model ablation in `artifacts/ablation/gemma-3-*_vs_*-abl.png`.
- Raw results + provenance: `artifacts/scaling/cleanliness.json`, `artifacts/ablation/*_vs_*-abl.json` (bf16 ladder) + the fp32 1B json, plus per-model ablation manifests.

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